

Question Paper

Exam Date & Time: 23-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026
SUBJECT: CSS 1001/CSS_1001 PROGRAMMING FOR PROBLEM SOLVING

Marks: 50

Duration: 180 mins.

Answer all the questions.

1A) Explain the role of an operating system in managing computer hardware. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

1B) Demonstrate through a flowchart and algorithm how two variables can be swapped without employing a third variable. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

1C) Design a C program that implements a digital clock using increment operators and loops. (5)
Sample output:
00:00:00
00:00:01
00:00:02
...
00:00:59
00:01:00

Course Outcome: "CO2" / Cognitive Level: "Application"

2A) Write a C program to print all odd numbers between 50 and 100 using a while loop. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

2B) i) Develop a C program to evaluate the expression (3)
result = (x++ * y-- + ++x / --y) % (x | y);
ii) considering initial values of x = 5, y = 3,
Evaluate the above expression step by step, considering precedence, associativity, and evaluation order.
(2+1 = 3 marks)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

2C) i) Analyze the working of the Bubble Sort algorithm with an example of sorting the array (5)
[7, 3, 5, 2, 8] in ascending order. Show the array contents after each pass and mention the number of comparisons and swaps made.
ii) Write a C program for the above situation.
(2+3 = 5 marks)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 3A) Explain the following string handling functions of C with proper syntax: strcpy(), strcat(). (2)
- Course Outcome: "CO3" / Cognitive Level: "Understand"
- 3B) Write a C program to remove all other characters in a string except the alphabet. Accept the string from the user. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 3C) Define recursive function. Write a C program to check whether the given number is Armstrong or not with the aid of a recursive function. (5)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- 4A) Write a C program that demonstrates two approaches to perform a mathematical operation, "incrementing a number by 5", using separate functions: one function that operates on a copy of the variable and another function that modifies the original variable directly. (3)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- 4B) Write a function to compute the sum of the first n cubes: $S(n) = 1^3 + 2^3 + \dots + n^3$ Print the sum in main() function. (3)
- Course Outcome: "CO4" / Cognitive Level: "Application"
- 4C) Write a C program to read M x N matrix in the main() function. Do the following tasks in function called process(). (4)
- i) Find the sum of rows and sum of columns.
 - ii) Replace all odd numbers in the matrix by binary equivalent of that number and even number by least number in the matrix.
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- 5A) Demonstrate different file opening modes and discuss putw() and feof() file functions with appropriate syntax. (3)
- Course Outcome: "CO5" / Cognitive Level: "Understand"
- 5B) Write a C program to find sum of all even numbers of a 1D array using a pointer to an array. (3)
- Course Outcome: "CO5" / Cognitive Level: "Application"
- 5C) Develop a C program to manage bank accounts using a structure. Your program should: (4)
- i) Allow the user to input the details (name, account number, account type, and balance) for multiple customers.
 - ii) Provide options to read customer details, display account information, and deposit an amount into a specific account.
 - iii) Implement error handling for invalid account numbers or deposit amounts.
 - iv) Ensure that the program continues to run until the user chooses to exit.
- Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSS_1001:PROGRAMMING FOR PROBLEM SOLVING

#	Course Outcome	Description	Marks
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1	1	CO 1	5
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	15
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSS_1001:PROGRAMMING FOR PROBLEM SOLVING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	4	10(20%)
3	Application	5	16(32%)
4	Analysis	6	24(48%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)